the controller is even called. This separation of concerns ensures that the controller can concentrate on the core logic of the operation, without having to worry about access control details. This also ensures that if you ever want to change how access control is performed, you can do this in the common filter rather than changing every single controller method.

To implement your access control rules, you need a filter that can determine whether the authenticated user has the appropriate permissions to perform a given operation on a given space. Rather than have one filter that tries to determine what operation is being performed by examining the request, you'll instead write a factory method that returns a new filter given details about the operation. You can then use this to create specific filters for each operation. The next listing shows how to implement this filter in your `UserController` class. You'll let the permission string be a regular expression to allow matching complex combinations of permissions.

Open UserController.java and add the method in listing 3.6 to the class underneath the other existing methods. The method takes as input the name of the HTTP method being performed and the permission required. If the HTTP method does not match then you skip validation for this operation, and let other filters handle it. Before you can enforce any access control rules, we must first ensure that the user is authenticated, so add an explicit call to the existing requireAuthentication filter first. Then you can look up the authenticated user in the user database and determine if they have the required permissions to perform this action, in